



Evaluation of spray applications of *Metarhizium anisopliae*, *Metarhizium brunneum* and *Beauveria bassiana* against larval winter ticks, *Dermacentor albipictus*

Cheryl Frank Sullivan¹ · Bruce L. Parker¹ · Agrin Davari¹ · Mi Rong Lee² · Jae Su Kim² · Margaret Skinner¹

Received: 8 July 2020 / Accepted: 16 September 2020 / Published online: 13 November 2020
© Springer Nature Switzerland AG 2020

Abstract

Dermacentor albipictus (Acari: Ixodidae), the winter tick, is a one-host tick that parasitizes large ungulates. They can dramatically affect moose, *Alces alces* (Artiodactyla: Cervidae), causing significant physiological and metabolic stress and mortality among heavily parasitized individuals. Entomopathogenic fungi in the genera *Metarhizium* (Hypocreales: Clavicipitaceae) and *Beauveria* (Hypocreales: Cordycipitaceae) are promising tick biological control agents. We examined the pathogenicity of experimental and commercially formulated isolates of *M. anisopliae*, *M. brunneum* and *B. bassiana* sprayed at concentrations of 10^6 , 10^7 and 10^8 conidia/mL against the larval stage of *D. albipictus* and assessed the efficacy of spraying the commercial product Met52®EC, containing *M. brunneum*, strain F52, under laboratory conditions. Results showed larval *D. albipictus* mortality was significantly higher and occurred earlier when treated with *M. anisopliae* and *M. brunneum* isolates compared to *B. bassiana* at 10^6 , 10^7 and 10^8 conidia/mL. Mortality was observed as early as 3 days in the *M. anisopliae* and *M. brunneum* treatments and after 6 days in the *B. bassiana* treatments. After 21 days, larval mortality ranged from 74–99% when ticks were treated with *M. anisopliae* and *M. brunneum* isolates at 10^6 , 10^7 and 10^8 and conidia/mL. In contrast, mortality of ticks treated with *B. bassiana* ranged from 30 to 64%. When larvae were treated with the commercial product Met52, mortality was ~45% after 3 days and ~96% after 9 days. These results demonstrate the effectiveness of *M. anisopliae* and *M. brunneum* against *D. albipictus*.

Keywords *Dermacentor albipictus* · Biological control · Biopesticide · Entomopathogenic fungi · *Metarhizium anisopliae* · *Metarhizium brunneum* · *Beauveria bassiana*

✉ Cheryl Frank Sullivan
cfrank@uvm.edu

Extended author information available on the last page of the article

Introduction

Microbial biopesticides containing entomopathogenic fungi for the biological control of ticks are an alternative to chemical pesticides (Beys-da-Silva et al. 2020; Fernandes et al. 2012). *Metarhizium anisopliae* sensu lato (s.l.) (Hypocreales: Clavicipitaceae) and *Beauveria bassiana* s.l. (Hypocreales: Cordycipitaceae) have been the focus of development into mycoinsecticides and mycoacaricides, worldwide. Several species and isolates are commercially available to manage a wide array of agricultural pests, but few are specifically registered for ticks (Arthurs and Dara 2019; Faria and Wraight 2007). *Metarhizium brunneum* Petch F52—formerly *Metarhizium anisopliae* (Metschnikoff) Sorokin strain F52 (Bischoff et al. 2009)—is produced as an oil-based emulsifiable concentrate and is labelled for ticks for applications to turf as a foliar spray or soil drench. Strain F52 has gained acceptance to manage the pathogen-vectoring blacklegged tick, *Ixodes scapularis* Say, with efficacy demonstrated under laboratory, residential and woodland conditions (Bharadwaj and Stafford 2010, 2012; Stafford and Allan 2010). Several formulations of *Beauveria bassiana* (Balsamo-Crivelli) Vuillemin are available and have been tested against ticks with promising efficacy. However, no *B. bassiana* products are labelled in the USA for specific use against ticks (Arthurs and Dara 2019). Both *B. bassiana* and *M. anisopliae* occur naturally in soils in the northeastern USA and have minimal non-target effects on other organisms making their use an attractive option for tick management (Brownbridge et al. 1993; Fischhoff et al. 2017; Parker et al. 2015; Zimmerman 2007a, b).

Dermacentor albipictus Packard (Acari: Ixodidae), the winter tick, is widely distributed across North America and northern Central America. It parasitizes many wildlife species such as moose, *Alces alces* (L.), elk, *Cervus canadensis* (Erxleben), and white-tailed deer, *Odocoileus virginianus* (Zimmermann). It also commonly infests livestock, such as horses, *Equus ferus caballus* L., and cattle, *Bos taurus* L. (Guzmán-Cornejo et al. 2016; Hays et al. 2019; Leo et al. 2014). *Dermacentor albipictus* is a one-host tick where larvae, nymphs and adults feed and moult on a single host (Bishopp and Wood 1913). This tick is especially problematic to moose causing epizootics (> 50% mortality), particularly in 11- to 12-month-old calves and decreased productivity in adult females. Although sporadic epizootics have occurred for the past century across the southern range of moose habitat, shorter winters associated with climate change have presumably contributed to increased epizootics over the past decade (Jones et al. 2019; Pekins 2020; Samuel 2004, 2007). Heavily infested calves die as a result of emaciation and metabolic imbalances caused by blood loss from excessive numbers of feeding ticks (Musante et al. 2007). An average of 47,371 ticks at a mean density of 2.9 ticks/cm² have been recently reported on moose in the northeastern USA (Jones et al. 2019). Peak feeding from adults occurs in March and April with the highest host mortality occurring over that time. During this period, engorged, mated females fall from their host to oviposit. In northern climates, *D. albipictus* produces one generation per year (Samuel 2004). Larvae hatch in summer and aggregate in large groups (averaging 1962 individuals) depending on the habitat conditions where females fell from their host (Aalangdong et al. 2001). They remain in a vulnerable, quiescent state to avoid desiccation (Yoder et al. 2016). They become active in autumn when they quest in large clumps on vegetation tips at a height of 50–190 cm (host torso height) off the ground (McPherson et al. 2000). They remain there until recruitment to their host or death occurs from inclement weather or starvation (Drew and Samuel 1985; Samuel 2004).

The free-living larval stage of this one-host tick species offers an ideal opportunity for their management prior to recruitment to their host. *Dermacentor albipictus* is susceptible

to numerous species and isolates of naturally-occurring fungi associated with moose and their habitats (Yoder et al. 2017, 2018, 2019a). Moose preferentially use patches of optimal habitat associated with timber harvest practices during spring when female ticks fall to the ground and the autumn when larvae quest for hosts (Healy et al. 2018). These areas have an abundance of ticks (Powers and Pekins 2020) and theoretically could be targeted for integrated tick management plans where biopesticides could be applied. We examined in vitro pathogenicity of experimental and commercial isolates of *M. anisopliae*, *M. brunneum* and *B. bassiana* sprayed at 10^6 , 10^7 and 10^8 conidia/mL against free-living, larval *D. albipictus*. We also assessed the efficacy of spraying the commercial formulation Met52®EC (Novozymes Biologicals, Franklinton, NC, USA), containing *M. brunneum* F52, at the label rates.

Materials and methods

Tick rearing

Individual rearing chambers were constructed with 30-mL (8 dram) polypropylene cups (Warmbuy; Shenzhen Zicome, Jiaxing City, China) to hold ticks for egg laying and larval eclosion. A 2-cm-diameter disc was removed from the bottom of the cup. Cups were retro-fitted with $150 \times 150 \mu\text{m}$ insect mesh (No-Thrips; BioQuip Products, Rancho Dominguez, CA, USA) adhered to the bottom for ventilation using a plastics bonding system (Loctite; Henkel, Rocky Hill, CT, USA). Engorged females were collected in April 2019 from a deceased moose calf in northeastern Vermont, USA. Ticks were placed, individually, into the rearing chambers then held within 750-mL, rectangular, polypropylene containers with vented lids (SureFresh; Greenbrier International, Chesapeake, VA, USA). A 60-mL (16 dram) cup (P200; Solo Cup, Lake Forest, IL, USA) filled 1/2 full of water was placed in each container to maintain humidity $> 90\%$ and vents were kept open. Containers were held at ambient temperature (ca. 20–25 °C) on a laboratory bench. A 46-cm (18 inch), 629 lumen light (GE; Jasco Products, Oklahoma City, OK, USA) was used to provide indirect light on a timer set for L16:D8 hours. Adult ticks were removed from the containers within 3 weeks of egg laying. To provide a standardized mixed culture of tick larvae for use in each set of experiments, eggs from three females were separated into equal sized groups and each group was then combined into individual batches.

Fungal isolate spray bioassays

Bioassay experiments were conducted to screen experimental and commercially produced isolates for pathogenicity against *D. albipictus* larvae (Table 1). Two sets of experiments were conducted. The first compared isolates of *M. anisopliae*, *M. brunneum* and *B. bassiana*. The second compared the *Metarhizium* spp. isolates with the addition of a third *M. anisopliae* isolate found to be pathogenic to *Haemaphysalis longicornis* Neumann (Lee et al. 2019) a recent invader to the USA (Rainey et al. 2018). Isolates were stored in 15% glycerol stock at -80°C in the University of Vermont Worldwide Collection of Entomopathogenic Fungi, Burlington, VT. Fungi were cultured for 14 days at ca. 25 °C on potato dextrose agar (Difco; Becton, Dickinson and Company, Franklin Lakes, NJ, USA) in 60×15 mm, polystyrene Petri dishes (Fisherbrand; Thermo Fisher Scientific, Waltham, MA, USA). Conidia were harvested by scraping the fungal mass with a spatula.

Table 1 Fungal isolates tested against *Dermacentor albipictus* larvae

Fungal species	Isolate code	Origin
<i>Metarhizium anisopliae</i>	ERL 1582	Soil from forest, Vermont, USA
	JEF 290	Soil from forest, South Korea
<i>Metarhizium brunneum</i> strain F52	F52	Met52® Granular, Novozymes Biologicals, USA
<i>Beauveria bassiana</i> strain GHA	GHA	BotaniGard®, Lam International, USA
<i>Beauveria bassiana</i>	ERL 836	Granular, FarmHannong, South Korea

Aqueous conidial suspensions were prepared in a 0.02% solution of organosilicone surfactant (Silwet L-77; Helena Agri-Enterprises, Collierville, TN, USA) and sterile distilled water (SDW). The suspensions were filtered through a fine mesh screen to remove mycelium. Conidia viability was determined before testing and suspensions were adjusted to obtain desired concentrations. Conidial concentrations were determined using a Neubauer Improved Brite-Line hemocytometer (Hausser Scientific, Horsham, PA, USA).

Tight-fit lid, 50×9 mm Petri dishes (Falcon; Corning Inc., Corning, NY, USA), lined with 47 mm, no. 1 Whatman filter paper (Cytiva, Marlborough, MA, USA) were used for the experiments. Ticks used in the experiments were 10 weeks old (for the first set of experiments) and 15 weeks old (second set). Ten actively crawling larvae were added to the center of each dish using a fine paintbrush (Artist's Loft fine 5/0; The Michaels, Irving, TX, USA). Ticks were treated by spraying with a hand-held, finger-tip sprayer from a 60-mL, travel-size bottle (Sprayco, Livonia, MI, USA) affixed to a 15-mL conical Fisherbrand centrifuge tube. Three concentrations of each fungal isolate (4×10^6 , 4×10^7 and 4×10^8 conidia/mL) and two controls: 0.02% Silwet+SDW solution (base control) and SDW only (untreated control) were prepared. Each treatment received 0.25 mL of spray from a height of 5 cm above the ticks to ensure delivery of the exposure concentrations (1×10^6 , 1×10^7 and 1×10^8 conidia/0.25 mL) to the treated area (19.6 cm^2) without supersaturating the filter paper. This was equivalent to rates of 5×10^4 , 5×10^5 and 5×10^6 conidia/cm². Dishes were placed in a white, plastic seed germination tray (54×29×6.5 cm) with a clear, plastic humidity dome and held on a benchtop in a quarantine room at ca. 25 °C and of L15:D9 photoperiod under an LED strip as described previously. A 60-mL Solo cup filled ¾ full of water was placed in the center of each tray to maintain humidity > 90%. Mortality was assessed every 3 days for 3 weeks for the first set of isolate experiments and every 3 days for 9 days for the second set. Each treatment had three replicates and the experiments were repeated 3× over three consecutive days. Ticks were evaluated microscopically at 60× magnification. Larvae were considered dead if they showed no movement when the dish was tapped and/or if fungal outgrowth was observed.

Met52 efficacy bioassay

The commercial product Met52 bioinsecticide containing *M. brunneum* F52 as the active ingredient was tested against 12-week-old larvae. Met52 contains 2×10^9 colony forming units/g (=conidia/g). The label recommendations for spray applications on turf for ticks is 59.14–88.72 mL/1000 ft² (≈ 0.64 – 0.93 mL/m^2). Three treatments were tested based on the range specified on the label, noted here as 'high' and 'low' with an SDW untreated control. The mean concentration of product delivered by following label instructions was estimated

by counting the number of conidia in the prepared suspensions using a hemocytometer. The mean number of conidia/mL was determined to be 1.6×10^7 conidia/mL (high rate) and 8.6×10^6 conidia/mL (low rate). Ticks were added to Petri dishes and treated using a hand-held atomizer as described previously. Each Petri dish received 0.25 mL of solution which equated to 4×10^6 conidia/dish (high rate) and 2×10^6 conidia/dish (low rate). This is the equivalent to rates of 2×10^5 and 1×10^5 conidia/cm², respectively. Conidia viability of the commercial product was determined prior to testing, using a method similar to that of Oliveira et al. (2015), after which application rates were adjusted to achieve desired conidia concentrations. Ticks were held and mortality was assessed as described above for the isolate bioassays. Mortality was assessed every 3 days for 9 days. Each treatment had five replicates and the experiment was repeated 3× over a period of three consecutive days.

Confirmation of fungal infection

Tick cadavers were surface sterilized at the end of the experiment to remove external contaminants. Ticks were cleaned in batches of five in a 6-well VWR tissue culture plate (VWR International, Radnor, PA, USA). They were soaked and vigorously stirred in 75% ethyl alcohol for 30 s, rinsed in a solution of 0.02% Silwet L-77 and SDW for 30 s, soaked in 1% bleach—SDW solution (6% sodium hypochlorite; The Clorox, Oakland, CA, USA), rinsed with 0.02% Silwet L-77—SDW solution for 30 s then followed by two 30-s rinses with SDW. Ticks were then placed on Difco potato dextrose agar supplemented with 0.02% chloramphenicol (ACROS Organics, Thermo Fisher Scientific) to reduce bacterial contamination in 100×15 mm Fisherbrand Petri dishes. Ticks were inspected for fungal growth after 1 week and target fungi confirmed 2–3 weeks later after conidiogenesis. Fungal colony features and conidia morphological characteristics were used for identification under 100× magnification (Humber 1997).

Data analysis

Mortality percentages of larvae were analyzed using a general linear model with repeated measures and univariate procedures, followed by Tukey's honestly significant difference (HSD) tests ($\alpha=0.05$). Data were not transformed and are reported in their original form. Analyses were conducted using IBM-SPSS v.26 (IBM, Armonk, NY, USA).

Results

Fungal isolate pathogenicity bioassays

Differences in the mortality of 10-week-old larvae were observed among *B. bassiana* (*Bb*), *M. anisopliae* (*Ma*) and *M. brunneum* (*Mb*) isolates when compared with the controls over the 21 days ($F_{78,504} = 11.27$, $P < 0.001$). No mortality was observed in the base control and 1% mortality was observed after 15 days in the untreated control treatment. Mortality was observed after 3 days in *M. anisopliae* and *M. brunneum* treatments and after 6 days in *B. bassiana*. After 21 days, mortality for the commercial isolate *Mb*-F52 ranged from 82 to 99%, and 74–86% for experimental isolate *Ma*-ERL1582. After 21 days, mortality was 30–41% for commercial isolate *Bb*-GHA and 52–64% for *Bb*-ERL836 (Fig. 1). Isolates applied at various conidial concentrations differed from

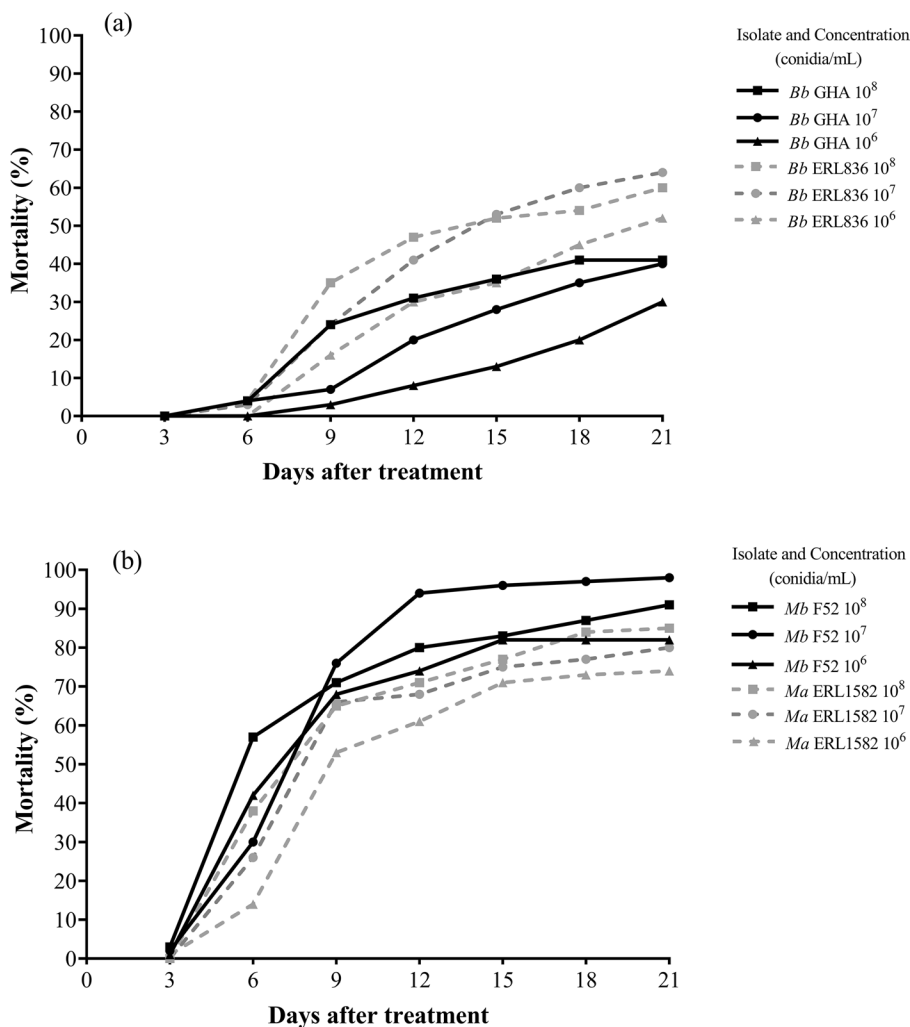


Fig. 1 Mean mortality (%) of 10-week-old *Dermacentor albipictus* larvae treated with conidia of **a** *Beauveria bassiana* (*Bb*) and **b** *Metarhizium anisopliae* (*Ma*) and *Metarhizium brunneum* (*Mb*) isolates at concentrations of 10^6 , 10^7 and 10^8 conidia/mL over 21 days

controls ($F_{13,84} = 36.64$, $P < 0.001$). All isolates and concentrations had greater mortality than control treatments ($P < 0.001$), except *Bb*-GHA where mortality was greater at 10^7 and 10^8 conidia/mL ($P < 0.006$) but not for 10^6 conidia/mL ($P = 0.28$). Larval mortality differed among isolates and concentrations over time ($F_{36,432} = 1.50$, $P = 0.034$). In general, larvae treated with *M. anisopliae* or *M. brunneum* had higher mortality than larvae treated with *B. bassiana* isolates at the same concentration. At 10^8 conidia/mL mortality did not differ between isolates *Mb*-F52 and *Ma*-ERL1582 ($P = 0.11$) but it did differ between the *Bb*-ERL2224 and *Bb*-ERL1722 isolates ($P = 0.009$) and between the *B. bassiana* and *Metarhizium* spp. isolates ($P < 0.001$). At 10^7 conidia/mL mortality was higher between the *Metarhizium* spp. isolates vs. the *B. bassiana* isolates ($P < 0.011$)

and there was no significant difference between the *B. bassiana* isolates ($P=0.080$) or the *Metarhizium* spp. isolates ($P=0.12$). A similar result was obtained for isolates at 10^6 conidia/mL. Concentration dependent virulence was observed for isolate *Mb*-F52 ($F_{12,108} = 2.92$, $P=0.002$) but not for *Ma*-ERL1582 ($F_{12,108} = 1.34$, $P=0.21$), *Bb*-GHA ($F_{12,108} = 1.35$, $P=0.20$) or *Bb*-ERL836 ($F_{12,108} = 1.24$, $P=0.27$) (Fig. 1).

In bioassays comparing the effect of three *Metarhizium* spp. isolates on 15-week-old larvae, mortality was greater among the fungal treatments compared to controls over the 9-day period ($F_{20,132} = 13.53$, $P<0.001$) (Table 2). There were no differences among isolates and concentrations throughout the 9-day experiment duration ($F_{8,108} = 1.62$, $P=0.16$). Mortality increased over time ($F_{2,108} = 600.16$, $P<0.001$) and differed among isolates ($F_{2,54} = 6.66$, $P=0.003$). Although there were no differences between *Ma*-JEF290, *Ma*-ERL1582 ($P=0.064$) and *Mb*-F52 ($P=0.40$), *Ma*-ERL1582 caused less overall mortality than *Mb*-F52 ($P=0.002$). A general concentration-dependent relationship was observed ($F_{2,54} = 45.3$, $P<0.001$) where mortality was higher at 10^8 than at 10^7 and 10^6 conidia/mL ($P<0.001$).

Met52 product bioassays

Differences in *D. albipictus* mortality were observed among treatments over the 9-day trial ($F_{4,72} = 38.19$, $P<0.001$) (Table 3). Larval mortality was greater for both rates compared to untreated control at each time period after application ($P<0.001$). Mortality differed over time ($F_{2,126} = 19.75$, $P<0.001$). Mortality also differed among the two rates at 3 days after application ($P=0.005$) but not at 6 days ($P=0.56$) and 9 days ($P=0.46$).

Table 2 Mean ($\pm$ SE) mortality (%) of 15-week-old *Dermacentor albipictus* larvae treated with two isolates of *Metarhizium anisopliae* (*Ma*) or with *Metarhizium brunneum* (*Mb*) at concentrations of 10^6 , 10^7 and 10^8 conidia/mL, a base control (0.02% Silwet + sterile distilled water [SDW] solution) and control (SDW) at 3, 6 and 9 days after treatment

Isolate	Treatment	Mortality after		
		3 days	6 days	9 days
<i>Ma</i> -JEF 290	Control	4.4 $\pm$ 2.4a	14.4 $\pm$ 2.9a	31.1 $\pm$ 4.2a
	Base Control	5.6 $\pm$ 2.4a	18.9 $\pm$ 2.6a	26.7 $\pm$ 3.7a
	10^8	73.3 $\pm$ 5.3c	100b	100b
	10^7	46.7 $\pm$ 9.1b	97.8 $\pm$ 2.2b	100b
	10^6	23.3 $\pm$ 4.1ab	91.1 $\pm$ 3.1b	98.9 $\pm$ 1.1b
<i>Ma</i> -ERL 1582	10^8	46.7 $\pm$ 6.5b	100b	100b
	10^7	34.4 $\pm$ 6.0b	98.9 $\pm$ 1.1b	100b
	10^6	24.4 $\pm$ 6.3ab	91.1 $\pm$ 5.4b	100b
<i>Mb</i> -F52	10^8	81.1 $\pm$ 5.1c	100b	100b
	10^7	46.7 $\pm$ 9.4b	97.8 $\pm$ 1.5b	100b
	10^6	33.3 $\pm$ 4.7b	92.2 $\pm$ 2.8b	100b

Means within a column followed by the same letter are not significantly different (Tukey's HSD: $P>0.05$)

Table 3 Mean ($\pm$ SE) mortality (%) of 12-week-old *Dermacentor albipictus* larvae after exposure to two rates of Met52 (2×10^5 and 1×10^5 conidia/cm²) and an untreated control (sterile distilled water) at 3, 6 and 9 days after application

Treatment	Mortality after		
	3 days	6 days	9 days
Control	2.0 $\pm$ 1.1c	7.3 $\pm$ 2.3b	13.3 $\pm$ 3.3b
1×10^5	26.0 $\pm$ 5.2b	88.0 $\pm$ 5.4a	94.0 $\pm$ 3.6a
2×10^5	43.3 $\pm$ 6.1a	92.7 $\pm$ 3.6a	98.0 $\pm$ 1.4a

Means within a column followed by the same letter are not significantly different (Tukey's HSD: $P > 0.05$)

Discussion

Dermacentor albipictus larvae were susceptible to increased mortality using spray applications of experimental isolates of *M. anisopliae* and commercial isolates of *M. brunneum* and *B. bassiana* within concentrations of 10^6 – 10^8 conidia/mL. The commercial biopesticide Met52 containing *M. brunneum* (formerly *M. anisopliae*) strain F52, when applied within label rates, also caused substantial mortality. Larvae at ages ranging from newly eclosed to 5 months have been previously shown to be susceptible to *M. brunneum* F52 isolated from Met52 when immersed in conidial solutions (Yoder et al. 2019b). Five-month-old larvae immersed in 1.6×10^7 conidia/mL had 50% survival after 3.7 days with none surviving by 15 days (Yoder et al. 2019a). We can now extend this knowledge to a range of concentrations effective at causing mortality to *D. albipictus* larvae.

Products containing strain F52 have been shown to be efficacious against other tick species, such as immature *I. scapularis*, when tested under laboratory, residential and woodland conditions (Bharadwaj and Stafford 2010, 2012; Stafford and Allan 2010). In laboratory experiments, spraying of nymphs with an emulsifiable concentrate of 2.6×10^5 conidia/cm² and keeping them on the treated surface for 30 min, resulted in 70.8% mortality after 4 weeks (Bharadwaj and Stafford 2012). Field spray applications of a strain F52 fungal biopesticide was applied at 3.2×10^5 conidia/cm² and 1.3×10^6 conidia/cm² for *I. scapularis* management in residential landscapes (Bharadwaj and Stafford 2010). After 5 weeks, an overall tick reduction of 53.2 and 73.8% was observed, respectively. In another field study, strain F52 applied at 2.5×10^{11} viable conidia/100 m² reduced *I. scapularis* nymphs by 55.6 and 84.6% in residential lawn and woodland landscapes, respectively (Stafford and Allan 2010). Given the high mortality observed in our results with larvae of *D. albipictus*, it's possible lower concentrations of strain F52 would provide significant larval suppression and would be worth investigating for application practicality. Nonetheless, our results in conjunction with the previously reported results support the potential of formulated *M. brunneum* F52 to manage *D. albipictus*. Field efficacy trials using this isolate formulated as a biopesticide against this species should be encouraged.

The efficacy of entomopathogenic fungi depend on several factors such as fungal isolate, environmental conditions, tick species and life stage (Fernandes et al. 2012; Ub and Narladkar 2018). In addition to *M. anisopliae*, *M. brunneum* and *B. bassiana*, *D. albipictus* larvae also vary in susceptibility to numerous other fungi genera found within the range of their natural habitat (Yoder et al. 2017, 2018, 2019a, b, c). Adult *I. scapularis* exposed to a commercial F52 strain formulation prepared at a concentration of 7.4×10^6 conidia/mL and applied at a rate of 2.6×10^5 conidia/cm² for 3, 30 and 300 s resulted in mortalities

of 33.3, 25, and 100%, respectively, after 6 weeks with mortality occurring as early as 1 week at temperatures of 5°C (Bharadwaj and Stafford 2012). This suggests fall applications could provide tick suppression until temperatures become uncondusive for sustained infection. Use of formulated fungus-based biopesticides for the management of this tick species should not be overlooked, especially where applications could be made in early autumn when larvae are most active (Yoder et al. 2017).

Entomopathogenic fungi have the potential to provide on-host tick biocontrol either through augmentation or natural occurrence. For example, when cattle were treated with oil-based formulations of *M. anisopliae* it suppressed an average of almost 50% of *Rhipicephalus microplus* (Canestrini) (Camargo et al. 2014) and 83% of *Rhipicephalus evertsi evertsi* Neumann and *Rhipicephalus (Boophilus) decoloratus* Koch (Kaaya et al. 2011). It was found that bull moose hair harbors several species of naturally occurring fungi that can be pathogenic to *D. albipictus* larvae. These fungi are presumably acquired by moose that frequent wallows (rut pits) where the fungi's natural environment has been significantly disturbed. After exposure, natural, fungus-based biocontrol could occur (Yoder et al. 2018, 2019c). Larval ticks treated with autumn applications of formulated, fungus-based biopesticides while questing, if then recruited to the host, may provide an additional source of fungal inoculum. This could provide an enhanced means of beneficial, on-host tick suppression if contact with enough fungal propagules occurs and on-host microclimate is conducive for fungal proliferation.

Virulence against ticks between species and isolates is largely dependent on application concentration. Treatment concentrations of 10^6 – 10^8 conidia/mL used in our experiments showed isolates of *M. anisopliae* and *M. brunneum* were superior in pathogenicity over the *B. bassiana* isolates. The experimental isolate *Ma*-JEF290 was highly pathogenic to soil-dwelling nymphs of *H. longicornis* (Lee et al. 2019) and was shown to be pathogenic against *D. albipictus*. In addition, an isolate from Vermont soil (*Ma*-ERL1582) was also promising. *Metarhizium anisopliae* has a high acaricidal activity from toxin production (destruxins) and a superior ability to penetrate the tick's external epicuticle layer as discussed in Lee et al. (2019) and Ub and Narladkar (2018). These experimental *M. anisopliae* isolates show potential for inclusion as biopesticides for ticks.

Beauveria bassiana isolates have been a major focus of biopesticide development and several formulations are commercially available for use in agricultural and forest settings (Arthurs and Dara 2019; Faria and Wraight 2007). Isolates of *M. anisopliae* cause higher morality in a shorter amount of time compared to *B. bassiana* when applied at similar concentrations. For example, *H. longicornis* nymphs were treated with *B. bassiana* and *M. anisopliae* isolates at concentrations of 10^6 – 10^8 conidia/mL with significantly greater mortality occurring for several *M. anisopliae* isolates over *B. bassiana* during a 14-day experimental period (Lee et al. 2019). *Beauveria bassiana* applied at concentrations of 10^7 and 10^9 conidia/mL needed 15 days or more to cause 100% mortality in larvae and nymphs of *H. longicornis* (Zhendong et al. 2019). Similar results were observed for unfed nymphs and larvae of *Rhipicephalus sanguineus* (s.l.) (Latreille), with isolates applied at 10^7 conidia/mL (Cafarchia et al. 2015). When *D. albipictus* larvae were immersed in *B. bassiana* solutions of 1.6×10^8 conidia/mL, mortality was as much as 91% within 10 days (Yoder et al. 2018). A commercial biopesticide containing the GHA isolate was sprayed at 9.9×10^{11} viable conidia/100 m² to lawn/woodland perimeters and woodlands in residential areas for *I. scapularis* management. Results showed a mean tick reduction of 74.5 and 55.2% in two consecutive years (Stafford and Allan 2010). Given the potential for *B. bassiana* to cause infection to *D. albipictus* and other tick species, the use of higher concentrations of *B. bassiana* within the specifications of commercial products should be investigated.

In conclusion, our results provide information on the potential of spray applications of experimental and commercially formulated entomopathogenic fungi to cause increased mortality to *D. albipictus* larvae. It demonstrates promise for use of biopesticide formulations containing *M. anisopliae* and *M. brunneum* against this tick species when used within label guidelines of commercially formulated products. Field efficacy of *M. anisopliae* and *M. brunneum* formulations should be investigated to broaden the use of fungus-based biopesticides against ticks. Use of and improvements to formulations containing *M. brunneum* and *M. anisopliae* should continue to be investigated until products become more available for tick management worldwide.

Acknowledgements This research was supported by the U.S. Geological Survey under Grant/Cooperative Agreement (No. G19AC00241), the U.S. Fish and Wildlife Service, Wildlife and Sport Fish Restoration Program, Wildlife Restoration Grants (No. 06120FY19522) by the Vermont Fish and Wildlife Department and the American Wildlife Conservation Foundation. We thank graduate students Jacob Debow and Joshua Blouin for field collection of ticks and the general collaboration with the Vermont Cooperative Fish and Wildlife Research Unit, Rubenstein School of Environment and Natural Resources. The use of trade or corporation names does not constitute an official endorsement by the University of Vermont, Jeonbuk National University or the supporting agencies.

Data availability <https://doi.org/10.6084/m9.figshare.12252722>.

Compliance with ethical standards

Conflict of Interest The authors declare that they have no conflict of interest.

References

- Aalangdong OI, Samuel WM, Shostak AW (2001) Off-host survival and reproductive success of adult female winter ticks, *Dermacentor albipictus* in seven habitat types of central Alberta. *J Ghana Sci Assoc* 3:109–116. <https://doi.org/10.4314/jgsa.v3i3.17773>
- Arthurs S, Dara SK (2019) Microbial biopesticides for invertebrate pests and their markets in the United States. *J Invertebr Pathol* 165:13–21. <https://doi.org/10.1016/j.jip.2018.01.008>
- Beys-da-Silva WO, Rosa RL, Berger M, Coutinho-Rodrigues CJB, Vainstein MH, Schrank A, Bittencourt VREP, Santi L (2020) Updating the application of *Metarhizium anisopliae* to control cattle tick *Rhipicephalus microplus* (Acari: Ixodidae). *Exp Parasitol* 208:107812. <https://doi.org/10.1016/j.exppara.2019.107812>
- Bharadwaj A, Stafford KC III (2010) Evaluation of *Metarhizium anisopliae* strain F52 (Hypocreales: Clavicipitaceae) for control of *Ixodes scapularis* (Acari: Ixodidae). *J Med Entomol* 47:862–867. <https://doi.org/10.1603/ME10020>
- Bharadwaj A, Stafford IIIKC (2012) Susceptibility of *Ixodes scapularis* (Acari: Ixodidae) to *Metarhizium brunneum* F52 (Hypocreales: Clavicipitaceae) using three exposure assays in the laboratory. *J Econ Entomol* 105:222–231. <https://doi.org/10.1603/EC11169>
- Bischoff JF, Rehner S, Humber RA (2009) A multilocus phylogeny of the *Metarhizium anisopliae* lineage. *Mycologia* 151:512–530. <https://doi.org/10.3852/07-202>
- Bishopp FC, Wood HP (1913) The biology of some North American ticks of the genus *Dermacentor*. *Parasitology* 6:153–187. <https://doi.org/10.1017/S0031182000003012>
- Brownbridge M, Humber RA, Parker BL, Skinner M (1993) Fungal entomopathogens recovered from Vermont forest soils. *Mycologia* 85:358–361. <https://doi.org/10.2307/3760695>
- Cafarchia C, Immediato D, Iatta R, Ramos RAN, Lia RP, Porretta D, Figueredo LA, Dantas-Torres F, Otranto D (2015) Native strains of *Beauveria bassiana* for the control of *Rhipicephalus sanguineus* sensu lato. *Parasit Vectors* 8:1–7. <https://doi.org/10.1186/s13071-015-0693-9>
- Camargo MG, Marciano AF, Sá FA, Perinotto WM, Quinelato S, Gôlo PS, Angelo IC, Prata MC, Bittencourt VR (2014) Commercial formulation of *Metarhizium anisopliae* for the control of *Rhipicephalus microplus* in a pen study. *Vet Parasitol* 205:271–276. <https://doi.org/10.1016/j.vetpar.2014.07.011>

- Drew ML, Samuel WM (1985) Factors affecting transmission of larval winter ticks, *Dermacentor albipictus* (Packard), to moose, *Alces alces* L., in Alberta, Canada. *J Wildl Dis* 21:274–282. <https://doi.org/10.7589/0090-3558-21.3.274>
- Faria MR, Wraight SP (2007) Mycoinsecticides and mycoacaricides: A comprehensive list with worldwide coverage and international classification of formulation types. *Biol Control* 43:237–256. <https://doi.org/10.1016/j.biocontrol.2007.08.001>
- Fernandes ÉKK, Bittencourt VREP, Roberts DW (2012) Perspectives on the potential of entomopathogenic fungi in biological control of ticks. *Exp Parasitol* 130:300–305. <https://doi.org/10.1016/j.exppara.2011.11.004>
- Fischhoff IR, Keesing F, Ostfeld RS (2017) The tick biocontrol agent *Metarhizium brunneum* (= *M. anisopliae*) (strain F52) does not reduce non-target arthropods. *PLoS One* 12:e0187675. <https://doi.org/10.1371/journal>
- Guzmán-Cornejo C, Robbins RG, Guglielmone AA, Montiel-Parra G, Rivas G, Pérez TM (2016) The *Dermacentor* (Acari, Ixodida, Ixodidae) of Mexico: hosts, geographical distribution and new records. *ZooKeys* 569:1–22. <https://doi.org/10.3897/zookeys.569.7221>
- Hays SR, Teel PD, Heath D, Starns NG, Moen R, Tolleson DR (2019) 125: Tick burden observed on cattle and sheep during winter season on the Edwards Plateau of Texas. *J Anim Sci* 97:53–54
- Healy CL, Pekins PJ, Kantar L, Congalton RG, Atallah S (2018) Selective habitat use by moose during critical periods in the winter tick life cycle. *Alces* 54:85–100
- Humber RA (1997) Identification of entomopathogenic fungi. In: Lacey LL (ed) *Manual of techniques in invertebrate pathology*, 1st edn. Academic Press, Amsterdam, pp 151–187
- Jones H, Pekins P, Kantar L, Sidor I, Ellingwood D, Lichtenwalner A, O’Neal M (2019) Mortality assessment of moose (*Alces alces*) calves during successive years of winter tick (*Dermacentor albipictus*) epizootics in New Hampshire and Maine (USA). *Can J Zool* 97:22–30. <https://doi.org/10.1139/cjz-2018-0140>
- Kaaya GP, Samish M, Hedimbi M, Gindin G, Glazer I (2011) Control of tick populations by spraying *Metarhizium anisopliae* conidia on cattle under field conditions. *Exp Appl Acarol* 55:273–281. <https://doi.org/10.1007/s10493-011-9471-3>
- Lee MR, Li D, Lee SJ, Kim JC, Kim S, Park SE, Baek S, Shin TY, Lee D, Kim JS (2019) Use of *Metarhizium anisopliae* s.l. to control soil-dwelling longhorned tick, *Haemaphysalis longicornis*. *J Invertebr Pathol* 166:1–8. <https://doi.org/10.1016/j.jip.2019>
- Leo SST, Pybus ST, Sperling MJ (2014) Origin of *Dermacentor albipictus* (Acari: Ixodidae) on elk in the Yukon, Canada. *J Wildl Dis* 50:544–551. <https://doi.org/10.7589/2013-03-078>
- McPherson M, Shostak AW, Samuel WM (2000) Climbing simulated vegetation to heights of ungulate hosts by larvae of *Dermacentor albipictus* (Acari: Ixodidae). *J Med Entomol* 37:114–120. <https://doi.org/10.1603/0022-2585-37.1.114>
- Musante AR, Pekins PJ, Scarpitti DL (2007) Metabolic impacts of winter tick infestations on moose calves. *Alces* 43:101–110
- Oliveira DGP, Pauli G, Mascarin GM, Delalibera I (2015) A protocol for determination of conidial viability of the fungal entomopathogens *Beauveria bassiana* and *Metarhizium anisopliae* from commercial products. *J Microbiol Methods* 119:44–52. <https://doi.org/10.1016/j.mimet.2015.09.021>
- Parker BL, Skinner M, Gouli SY, Gouli VV, Tobi D, Kim JS (2015) Persistence of *Beauveria bassiana* sensu lato and *Metarhizium anisopliae* sensu lato in Vermont (USA) forest soil. *Biocontrol Sci Technol* 25:768–788. <https://doi.org/10.1080/09583157.2015.1016895>
- Pekins PJ (2020) Metabolic and population effects of winter tick infestations on moose: unique evolutionary circumstances? *Front Ecol Evol* 8:176. <https://doi.org/10.3389/fevo.2020.00176>
- Powers BI, Pekins PJ (2020) Abundance of winter ticks (*Dermacentor albipictus*) in two regenerating forest habitats in New Hampshire, USA. *Alces* 56:1–13
- Rainey T, Occi JL, Robbins G, Egizi RA (2018) Discovery of *Haemaphysalis longicornis* (Ixodida: Ixodidae) parasitizing a sheep in New Jersey, United States. *J Med Entomol* 55:757–759. <https://doi.org/10.1093/jme/tjy006>
- Samuel B (2004) White as a ghost: Winter ticks and moose. *Federation of Alberta Naturalists*, Edmonton
- Samuel WM (2007) Factors affecting epizootics of winter ticks and mortality of moose. *Alces* 43:39–48
- Stafford IIIKC, Allan SA (2010) Field applications of entomopathogenic fungi *Beauveria bassiana* and *Metarhizium anisopliae* F52 (Hypocreales: Clavicipitaceae) for the control of *Ixodes scapularis* (Acari: Ixodidae). *J Med Entomol* 47:1107–1115. <https://doi.org/10.1603/ME10019>
- Ub GR, Narladkar BW (2018) Role of entomopathogenic fungi in tick control: A review. *J Entomol Zool Stud* 6:1265–1269

- Yoder JA, Pekins PJ, Jones HF, Nelson BW, Lorenz AL, Jajack A (2016) Water balance attributes for off-host survival in larvae of the winter tick (*Dermacentor albipictus*; Acari: Ixodidae) from wild moose. *Int J Acarol* 42:26–33. <https://doi.org/10.1080/01647954.2015.1113310>
- Yoder JA, Pekins PJ, Nelson BW, Randazzo CR, Siemon BP (2017) Susceptibility of winter tick larvae and eggs to entomopathogenic fungi—*Beauveria bassiana*, *Beauveria caledonica*, *Metarhizium anisopliae* and *Scopulariopsis brevicaulis*. *Alces* 53:41–51
- Yoder JA, Dobrothka CJ, Fisher KA, LeBarge AP, Pekins PJ, McLellan M (2018) Entomopathogenic fungi of the winter tick in moose wallows: a possible bio-control for adult moose. *Alces* 54:55–70
- Yoder JA, Krieger M, Oakley M, Trotter J, Schmelzer P, Niksic A, Rodell BM, Klever LA (2019) Growth characteristics and pathogenic consequences of predominant entomopathogenic Yukon soil fungi *Mortierella alpina* and *Penicillium expansum*, and effectiveness of Met52®, against larvae of the winter tick, *Dermacentor albipictus*. *Stud Fungi* 4:94–103. <https://doi.org/10.5943/sif/4/1/13>
- Yoder JA, Rodell BM, Klever LA, Dobrotka CJ, Pekins PJ (2019b) Vertical transmission of the entomopathogenic soil fungus *Scopulariopsis brevicaulis* as a contaminant of eggs in the winter tick, *Dermacentor albipictus*, collected from calf moose (New Hampshire, USA). *Mycology* 10:174–181. <https://doi.org/10.1080/21501203.2019.1600062>
- Yoder JA, Pekins PJ, Dobrotka CJ, Fisher KA, Randazzo CR, Kantar L, McLellan S, O'Neal M (2019c) Enhancement of fur and skin from bull moose with additional entomopathogenic fungi that offer increased protection against damage from winter ticks (*Dermacentor albipictus*; Acari: Ixodidae). *Int J Acarol* 45:97–105. <https://doi.org/10.1080/01647954.2018.1554700>
- Zhendong H, Guangfu Y, Zhong Z, Ruiling Z (2019) Phylogenetic relationships and effectiveness of four *Beauveria bassiana* sensu lato strains for control of *Haemaphysalis longicornis* (Acari: Ixodidae). *Exp Appl Acarol* 77:83–92. <https://doi.org/10.1007/s10493-018-0329-9>
- Zimmerman G (2007a) Review on safety of the entomopathogenic fungi *Beauveria bassiana* and *Beauveria brongniartii*. *Biocontrol Sci Tech* 17:553–596. <https://doi.org/10.1080/09583150701309006>
- Zimmerman G (2007b) Review on safety of the entomopathogenic fungus *Metarhizium anisopliae*. *Biocontrol Sci Tech* 17:879–920. <https://doi.org/10.1080/09583150701593963>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Affiliations

Cheryl Frank Sullivan¹  · Bruce L. Parker¹ · Agrin Davari¹ · Mi Rong Lee² · Jae Su Kim² · Margaret Skinner¹

Bruce L. Parker
bparker@uvm.edu

Agrin Davari
adavari@uvm.edu

Mi Rong Lee
mimilee248@naver.com

Jae Su Kim
jskim10@jbnu.ac.kr

Margaret Skinner
mskinner@uvm.edu

¹ Entomology Research Laboratory, University of Vermont, Burlington, VT, USA

² Department of Agricultural Biology, Jeonbuk National University, Jeonju, Republic of Korea